Reverse Words In String

AlgoExpert · Medium · Strings

The problem

- Reverse the order of space-separated words, preserving all whitespace runs.

Intuition

- Reverse the order of words while keeping whitespace runs intact.

Brute force vs optimal

- Split + reverse + join (library) → O(n)
- Manual scan from the end collecting tokens → O(n)
- In-place: reverse all, then reverse each word

Approach

- 1. Tokenise into words and spaces
- 2. Reverse the token list
- 3. Join back

Worked example

```
'AlgoExpert is the best!'
```

Input

Worked example



Tokens: words and space runs

Worked example




```
'best! the is AlgoExpert'
```

Reverse token order

Java solution

```java
public class ReverseWordsInString {
    public static String reverseWordsInString(String s) {
        StringBuilder out = new StringBuilder();
        int end = s.length();
        for (int i = s.length() - 1; i >= -1; i--) {
            if (i == -1 || (s.charAt(i) == ' ') != (s.charAt(end - 1) == ' ')) {
                out.append(s, i + 1, end);
                end = i + 1;
            }
            if (end == 0) break;
        }
        return out.toString();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Edge cases & pitfalls

- Multiple spaces preserved
- Leading/trailing spaces
- Empty string

Interview tips & follow-ups

- Mention the in-place two-reversal trick

Recap

- Reverse the order of words while keeping whitespace runs intact.
- Complexity: O(n)